MIT
MIVA OPEN UNIVERSITY

School of Post Graduate Studies

Master of Information Technology (MIT)

40 marks

2026/2027 Academic Session
Continuous Assessment

MIT 8301 - Machine Learning and AI Fundamentals (Virtual lab)

Submission Guidelines

Note that your submission should be in one PDF document and it should:

- Include all the code base
- Include screenshots of output
- Any other indicated requirement in this document

Instruction: Use the following dataset

Dataset link: https://www.kaggle.com/datasets/uciml/german-credit

Description: This dataset contains information about loan applicants, including their financial history, loan amount, and whether they defaulted on the loan

Task

Instruction

1. Load the dataset and perform exploratory data analysis (EDA) to understand its characteristics.
2. Handle missing values, outliers, and categorical variables appropriately.
3. Split the data into training and testing sets.

Model Selection and Training

For classification (churn prediction, credit risk assessment):

Instruction

Use the following algorithms; Logistic Regression, Support Vector Machines, Naive Bayes to build models Logistic Regression - from scratch

Hyperparameter Tuning

Instruction

1. Tune the hyperparameters of the chosen models using techniques like Grid Search or Randomized Search to optimize their performance.
2. Explain the importance of hyperparameter tuning and how it affects model performance.

Model Evaluation and Comparison

Compare the models and analyze their strengths and weaknesses for the given problem using Accuracy, Precision, Recall, F1-score, ROC AUC.

Feature Importance and Interpretation

For the best-performing model, analyze the feature importance to understand which factors are most influential in making predictions.

Discuss how these insights can be used to inform decision-making in the given scenario

Rubric for MIT 8301_Virtual Lab Exercise (Total: 40 points)

1. Data Loading & EDA – 6 points

Excellent (6 pts): Correctly loads dataset, displays basic info (shape, columns, data types), performs descriptive statistics, checks target balance, and identifies categorical vs numerical features. Includes insightful visualizations (histograms, boxplots, etc.).

Satisfactory (3–5 pts): Loads data but missing some EDA steps (e.g., no target balance check, no visualizations).

Needs improvement (0–2 pts): No EDA or major omissions (e.g., no data inspection, no identification of feature types).

2. Handling Missing Values & Outliers – 5 points

Excellent (5 pts): Identifies and correctly handles missing values (imputation or removal). Detects outliers using IQR or visual methods and applies appropriate capping/transformation. Justification provided.

Satisfactory (2–4 pts): Handles missing values but not outliers, or vice versa. Limited justification.

Needs improvement (0–1 pt): Ignores missing values and outliers, leading to incorrect data for modeling.

3. Categorical Encoding & Scaling – 5 points

Excellent (5 pts): Correctly applies one-hot or label encoding based on feature nature. Splits data and scales numerical features using StandardScaler or MinMaxScaler. No data leakage.

Satisfactory (2–4 pts): Encoding or scaling is partially correct (e.g., scales before split, or uses wrong encoding type).

Needs improvement (0–1 pt): No encoding or scaling, or uses ordinal encoding for nominal categories without reason.

4. Logistic Regression from Scratch – 8 points

Excellent (8 pts): Implements a fully functional logistic regression class with sigmoid, cost function, gradient descent, and prediction methods. Code is clean, vectorized, and converges. Results reasonably match scikit-learn's version.

Satisfactory (4–7 pts): Implementation works but has minor bugs (e.g., no bias term, wrong stopping condition) or performance issues.

Needs improvement (0–3 pts): Code does not run, fails to converge, or misses key components (e.g., no gradient descent).

5. Model Training & Hyperparameter Tuning – 6 points

Excellent (6 pts): Trains at least three models (Logistic Regression, SVM, Naive Bayes). Performs grid or random search on relevant hyperparameters (e.g., C, kernel) with cross-validation. Explains why tuning is important.

Satisfactory (3–5 pts): Trains models but tuning is incomplete (e.g., only default parameters, or only one model tuned).

Needs improvement (0–2 pts): No tuning, or uses test data for tuning. No explanation of importance.

6. Model Evaluation & Comparison – 6 points

Excellent (6 pts): Computes all requested metrics (accuracy, precision, recall, F1-score, ROC AUC) on the test set. Compares models in a clear table or visualization, discussing strengths and weaknesses (e.g., imbalance, interpretability).

Satisfactory (3–5 pts): Computes some metrics but missing others; comparison is superficial.

Needs improvement (0–2 pts): Uses only accuracy on imbalanced data, or no comparison.

7. Feature Importance & Interpretation – 4 points

Excellent (4 pts): Extracts coefficients from the best model (e.g., logistic regression) and provides a ranked list of top influential features. Discusses real-world impact for credit risk decisions (e.g., policy, pricing).

Satisfactory (2–3 pts): Mentions feature importance but no ranking or shallow discussion.

Needs improvement (0–1 pt): No interpretation, or uses a black-box model without any explainability.